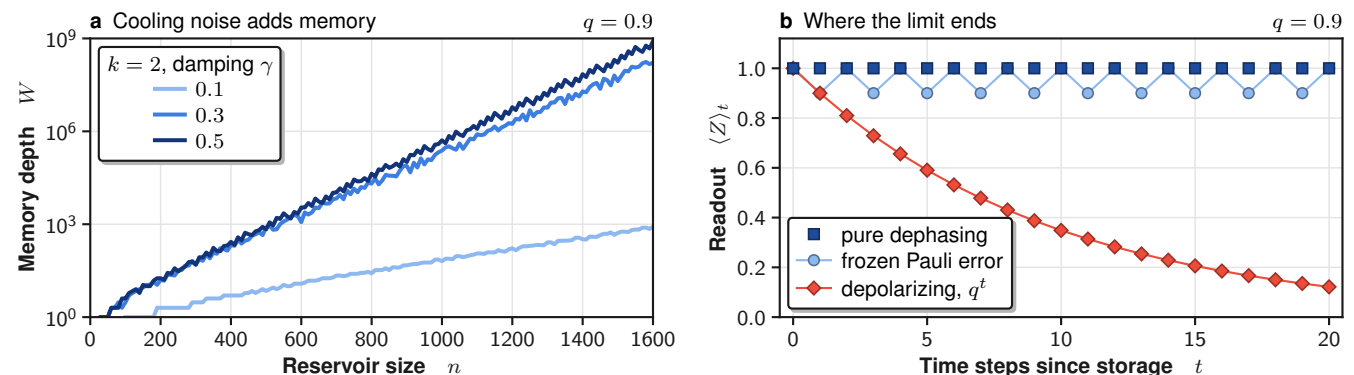

a Cooling noise adds memory
$q = 0.9$
$k = 2$, damping γ
0.1
0.3
0.5
Memory depth W
10^9
10^6
10^3
10^0
0
200
400
600
800
1000
1200
1400
1600
Reservoir size n
b Where the limit ends
$q = 0.9$
Readout $\langle Z \rangle_t$
1.0
0.8
0.6
0.4
0.2
0.0
pure dephasing
frozen Pauli error
depolarizing, q^t
0
5
10
15
20
Time steps since storage t